

Answer to the ques. no. 1.

(i)

Given,

$$\text{frequency} = 50 \text{ kHz}$$

$$\therefore \text{Time period, } T = \frac{1}{50} \text{ ms.}$$

$$\text{Total possible count} = (2^8 - 1).$$

$$\therefore \text{Total time} = \frac{1}{50} \times (2^8 - 1) \text{ ms.}$$

$$\text{Ramp slope} = 1.953 \text{ V/ms}$$

$$\begin{aligned} \therefore \text{Max voltage} &= \frac{1}{50} \times (2^8 - 1) \times 1.953 \\ &= 9.9603 \text{ V.} \end{aligned}$$

$$\therefore \text{Quantization step size} = \frac{V_{\text{max}} - V_{\text{min}}}{L}$$

$$= \frac{9.9603}{2^8}$$

$$= 0.0389 \text{ V}$$

$$\approx 39 \text{ mV.}$$



(ii) if all of them

From (i),

$$V_{\max} = 9.9603 \text{ V.}$$

No. of counts for 9.9603 V is  $2^8 - 1$ .

$$\therefore \frac{9.9603}{3.91} = \frac{2^8 - 1}{x} \times 3.91$$
$$= 100.10$$

In 8-bit binary: 01100100

(iii)

$$\text{Total counts} = 2^8 - 1$$

$$\text{Time period, } T = \frac{1}{50} \text{ ms.}$$

$$\therefore \text{Maximum conversion time} = (2^8 - 1) \times \frac{1}{50}$$
$$= 5.1 \text{ ms.}$$



Answer to the ques. no - 2

(i)

Given,

Frequency = 100 kHz

Time = 40.96 ms

In 1 sec. no. of clock pulse =  $100 \times 10^3$

$$\therefore 40.96 \text{ ms} \times 100 \times 10^3 \times 40.96 \times 10^{-3} = 4096$$

$\therefore$  number of counts accumulated during integration phase is 4006.

(ii)

Maximum count = 4096.

$$\therefore \text{no. of bits in the counter} = \log_2(4096) = 12.$$

(ii)

no-of counts for reaching 5 volt = 4095

$$2.41 \times \frac{4096 \times 2.41}{5} = 1974.272$$

$$\therefore \text{Required time} = T \times 1974.272$$

$$= \frac{1}{100 \times 10^3} \times 1974.272$$

$$= 0.019745$$

$$= 19.74 \text{ ms.}$$



(iv) Total counts

No. of counts = 1974.27  $\approx$  1974.

In binary = 0111 1011 0110.

Answer to the ques. no-3

Ques (i)

For single slope:

$V_{ref} = 5V$ .

Maximum conversion time, =  $\frac{2^N - 1}{\text{Frequency}} = \frac{2^N - 1}{106} = 4.090 \text{ ms}$

Time required for 4.5 V =  $\frac{4.090}{5} \times 4.5 = 3.685 \times 10^{-3}$   
 $= 3.685 \text{ ms}$ .

For dual slope:

Time required =  $(4.090 + 3.685) \text{ ms}$

$= 7.775 \text{ ms}$ .



(ii)

For single slope,

$$\text{No. of counts} = 3.6855 \times \text{frequency} \times 10^{-3}$$

$$= 3.6855 \times 10^{-3} \times 10^6$$

$$= 3685.5$$

$$\approx 3686$$

$\therefore$  Digital output code is 111001100110.

For dual slope:

$$\text{No. of counts} = 3686$$

Binary: 111001100110

(iii)

Dual slope ADC is generally more suitable for high accuracy or noisy environments, because:

☐ It integrates the input for a fixed time, so random and high frequency noise is averaged out.

☐ If the integration time is set to match the main cycle,



it cancels power-line interference.

□ The result depends mainly on the stable reference voltage during de-integration, not on the  $RC$  time constant of the integrator.

In contrast, the single-slope ADE is faster, but its output strongly depends on the  $RC$  values of the ramp circuit. Any variation in  $R$  or  $C$ , or noise in the ramp, introduces errors.

Therefore, dual-slope ADEs are more suitable for high-accuracy measurements in noisy environments.